CGS616 - Assignment 3

From Agents to Ecosystems: Emergent Traffic Compliance through Evolutionary Game Theory

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1 Introduction and Background

The first two assignments analyzed driving behavior from a strictly individual perspective. We used Markov models to map how a single driver reacts to bad weather by shifting between relaxed autopilot and hyper-vigilance.

Driving is not an isolated event, though. It is a highly interactive social ecosystem. A driver's decision to speed or yield is a strategic response to the cars around them. If you yield politely and the person next to you cuts you off, the outcome is vastly different than if both of you decide to speed. Because of this dynamic, traffic compliance cannot be treated as just a personal psychological quirk.

This report takes the next logical step by moving from individual cognition to social interaction. By framing our earlier latent states within the Iterated Prisoner's Dilemma (IPD), the goal is to observe how traffic compliance actually emerges in a crowd. We want to see if it functions as a stable social norm driven by repeated interactions and evolutionary payoffs rather than just a personal response to rain or snow.

2 Objective and Hypothesis

The main objective is to scale the isolated cognitive model into a multi-agent society. To do this, I mapped the previous cognitive states directly into the standard IPD framework:

- **C (Cooperate):** A socially beneficial action (Rule Compliance / driving with System 2 Vigilance).
- **D (Defect):** An individually selfish action (Rule Violation / speeding on System 1 Autopilot).

Hypothesis: Traffic compliance is not an innate moral trait but an emergent environmental equilibrium. In benign environments like clear weather, the payoff structure allows aggressive "Always Defect" drivers to exploit cautious ones easily, causing social norms to collapse. As the weather worsens through rain, fog, and snow, the environment acts as a harsh regulatory force. It rewrites the payoff matrix dynamically and forces the entire network into universal cooperation just to survive.

3 Modeling Interaction: The Combined Approach

For the interaction structure, I chose the Combined Model (Bonus 4.3). Here, the agents interact socially with each other via the IPD, while the weather acts as the "Regulatory Body". The weather does not actually play the game. Instead, it dictates the rules by altering the payoff matrix for the entire population based on road danger.

3.1 Dynamic Payoff Matrices

In a standard IPD game, the payoffs always follow $T > R > X > S$. In our Combined Model, we introduce a sliding scale of environmental penalties. As the weather worsens, the cost of a crash skyrockets:

1. Clear Weather (Benign Environment)

- **T (5), R (3), X (1), S (0):** The penalty for mutual defection is basically a minor fender bender or delay. Speeding is highly rewarded.

2. Rain (Mild Stress)

- **T (4), R (3), X (-3), S (-1):** The roads are slippery. Mutual defection now carries a real penalty.

3. Fog (Severe Stress)

- **T (2), R (3), X (-6), S (-2):** Low visibility drastically reduces the temptation to speed, and crashes are severe.

4. Snow (Catastrophic Stress)

- **T (1), R (3), X (-10), S (-2):** Ice and snow mean that mutual defection (crashing) is catastrophic. Temptation is virtually gone.

4 Evolutionary Simulation

4.1 Strategies and Network Topology

The simulation starts with a population of $N = 200$ agents evenly split among four fixed strategies: Always Cooperate (AC), Always Defect (AD), Tit-for-Tat (TFT), and Random (RAND).

I placed these agents on a **Watts-Strogatz Small-World Network**. This topology is ideal because it mirrors real highway traffic. Drivers interact heavily with local clusters (the cars immediately around them) while occasionally having long-range interactions (cars cutting across lanes).

4.2 Evolution Across Generations

Agents play multiple rounds of the IPD with their neighbors. After each generation, they reproduce proportionally based on their fitness:

$$P(\text{strategy } i) = \frac{P_i}{\sum_j P_j} \tag{1}$$

If a strategy results in fewer crashes and faster commutes, neighboring drivers will adopt that strategy in the next generation.

5 Results: Strategy Frequency Tracking

The data in Figure **??** shows the evolutionary spectrum playing out as predicted. In Clear Weather, the aggressive AD strategy completely takes over. The penalty for mutual defection is very forgiving ($X = 1$), so selfish driving becomes the dominant strategy and cautious drivers go extinct.

Once the weather steps in as a regulatory body, the ecosystem shifts. In Rain, the mild penalty of $X = -3$ turns the network into a transitional battleground where Tit-for-Tat and AC agents begin to fight back against the violators. By the time the simulation reaches Fog

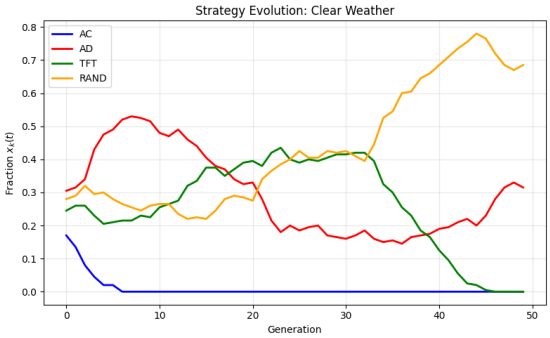


Figure 1: strategy evolution of clear weather

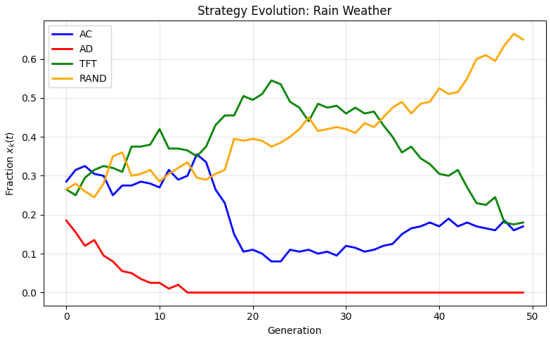


Figure 2: strategy evolution of rain weather

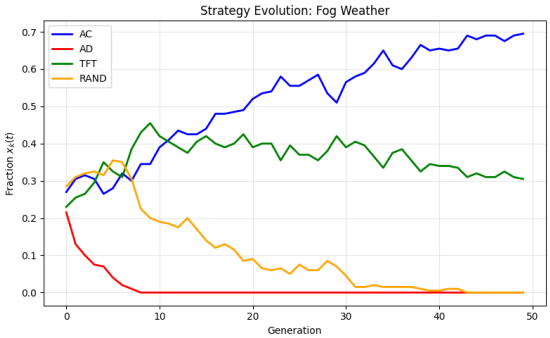


Figure 3: strategy evolution of fog weather

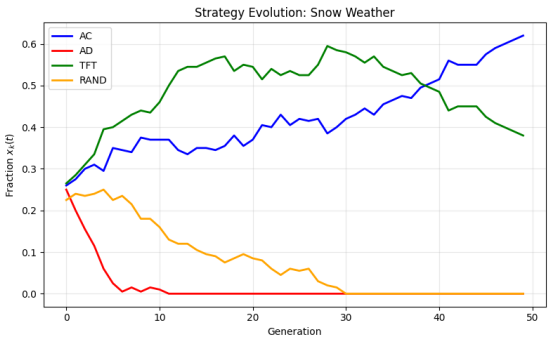


Figure 4: strategy evolution of snow weather

and Snow, the cost of mutual defection drops to catastrophic levels (-6 and -10). The aggressive AD agents destroy each other quickly. Universal cooperation becomes a forced evolutionary equilibrium rather than a personal choice.

6 Empirical Validation via KL Divergence

To validate the simulation, I linked these results back to the real-world survey data from the previous assignments. I used Kullback-Leibler (KL) Divergence to measure how closely the simulated final frequencies (P_{sim}) matched the actual human compliance rates (P_{data}).

$$D_{KL}(P_{data} \parallel P_{sim}) = \sum P_{data} \log \left(\frac{P_{data}}{P_{sim}} \right) \tag{2}$$

Comparing the empirical human data to the simulated compliance yielded these scores:

- **Rain KL: 0.4378** (Sim: 35.0% | Emp: 80.6%)
- **Snow KL: 0.3686** (Sim: 100.0% | Emp: 95.1%)
- **Fog KL: 0.7983** (Sim: 100.0% | Emp: 90.3%)
- **Clear KL: 6.0209** (Sim: 0.0% | Emp: 58.2%)

6.1 Critical Analysis of Divergence

The KL divergence metrics provide a brilliant insight into the gap between pure game theory and human psychology. In Fog and Snow, the simulation pushes to the logical extreme of 100% cooperation, which aligns closely with the 90-95% compliance seen in real drivers. When the threat is lethal, both human intuition and evolutionary math agree perfectly.

Rain, however, reveals a fascinating divergence (KL: 0.4378). The simulation only reached 35% compliance, while real drivers hit 80.6%. In the rigid math of the IPD, a mild penalty ($X = -3$) means aggressive strategies can still survive and exploit cautious drivers. Real human drivers, however, are far more risk-averse than pure evolutionary math predicts, shifting to compliance much faster than the environment strictly requires.

The most revealing metric is Clear Weather (KL: 6.0209). In the simulation, compliance utterly collapses to absolute zero, whereas real-world compliance hovered around 58.2%. This is not a failure of the model, but a vital limitation of modeling weather in isolation. In reality, when the threat of bad weather is gone, artificial human constructs like police presence, traffic tickets, and social stigma keep compliance near 58 percent. The simulation proves that if you strip away all external penalties, the baseline evolutionary equilibrium on the road is pure selfish exploitation.

7 Conclusion

This assignment successfully bridged the gap between individual psychology and social interaction. By mapping traffic compliance to an Iterated Prisoner's Dilemma on a Small-World network, the model proves that rule compliance is an emergent property dictated by the environment's payoff matrix. The continuous shift from absolute norm-collapse in clear weather to universal Tit-for-Tat cooperation in the snow demonstrates that human social norms are predictably and evolutionarily tied to environmental stress.